

MEDIMATCH · NATIONAL REDISTRIBUTION COMMAND

See surplus. Detect need. Coordinate impact.

Leveraging geospatial intelligence
for equitable medical-supply redistribution

Lesnar Gitonga · USIU-Africa · 23 June 2026

Conference demo · synthetic inventory · no patient records

The problem is not only scarcity. It is disconnected visibility.

Facilities can hold usable surplus while others face urgent shortfalls. MediMatch makes both sides visible in one operating picture.

56.3%

face weekly or monthly stockouts

82.8%

rate manual methods ineffective or slow

57.8%

report wastage of valid supplies

89.1%

willing to pilot the system

A supply gap becomes a coordination question

Which available source is close enough, sufficiently stocked, verified, product-compatible—and practical to move now?

57%

KEMSA national order-fill rate,
as of mid-2025 (per the abstract)

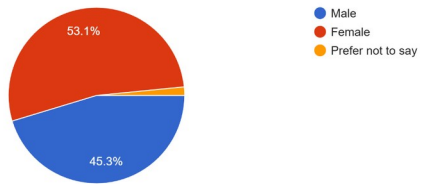
The baseline is historical; the deck does not present it as the current 2026 rate.

Who we surveyed

64 healthcare professionals · Nairobi County · 80% response rate · unedited Google Forms output

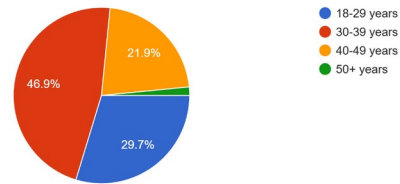
A frontline sample — the clinicians and staff who manage stock every day.

Q1. What is your gender?
64 responses



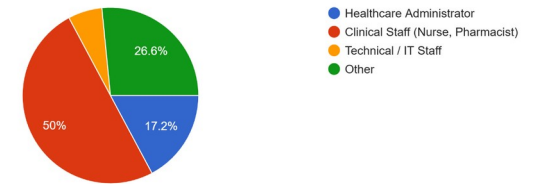
Q1 · Gender — slight female majority

Q2. What is your age bracket?
64 responses



Q2 · Age — majority under 40

Q3. What is your primary role at this facility?
64 responses



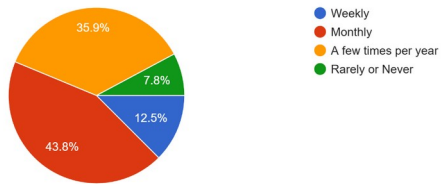
Q3 · Role — clinical staff majority

The current system, in their words

Questions 4–6 · frequency of stockouts, how surplus is handled, and how they rate today's methods

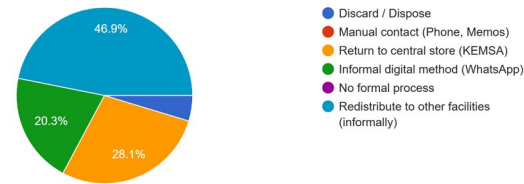
Frequent stockouts, ad-hoc handling, and locating methods they rate as ineffective or slow.

Q4. How often does your facility experience stockouts of essential items?
64 responses



Q4 · 56.3% weekly or monthly stockouts

Q5. What is your facility's current method for handling surplus (near-expiry) supplies?
64 responses



Q5 · Surplus handled ad-hoc

Q6. How would you rate the effectiveness of your current method for locating emergency supplies from nearby facilities?
64 responses



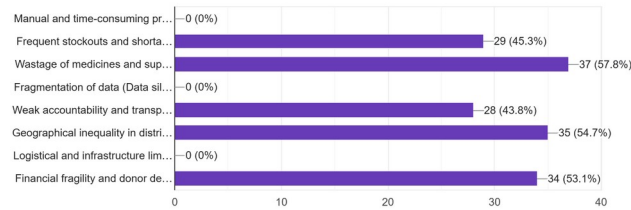
Q6 · 82.8% ineffective or slow

Challenges, value & readiness

Questions 7–9 · the main system challenges, the benefits they value, and willingness to pilot

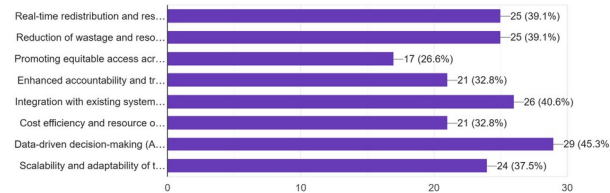
Wastage tops the operational challenges — and readiness to pilot MediMatch is high.

Q7. Which of the following do you consider the main challenges of the current supply system?
64 responses



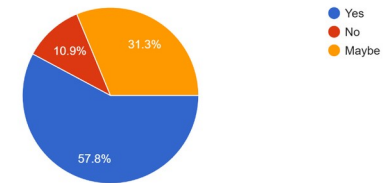
Q7 · 57.8% cite wastage of supplies

Q8. Which of the following benefits would be most valuable if a new digital platform (MediMatch) were implemented?
64 responses



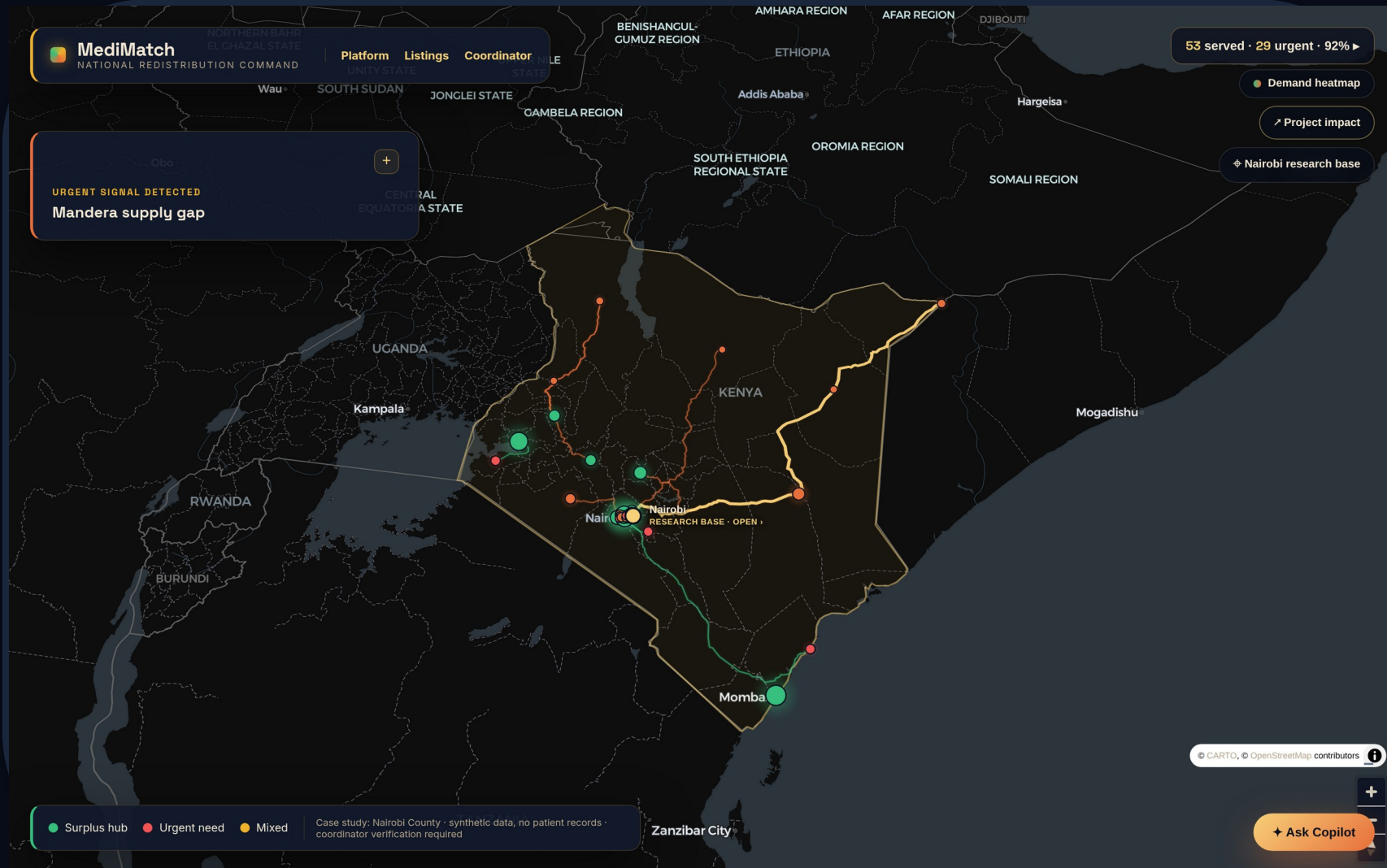
Q8 · Most-valued benefits

Q9. Would you recommend your facility explore piloting the MediMatch platform?
64 responses

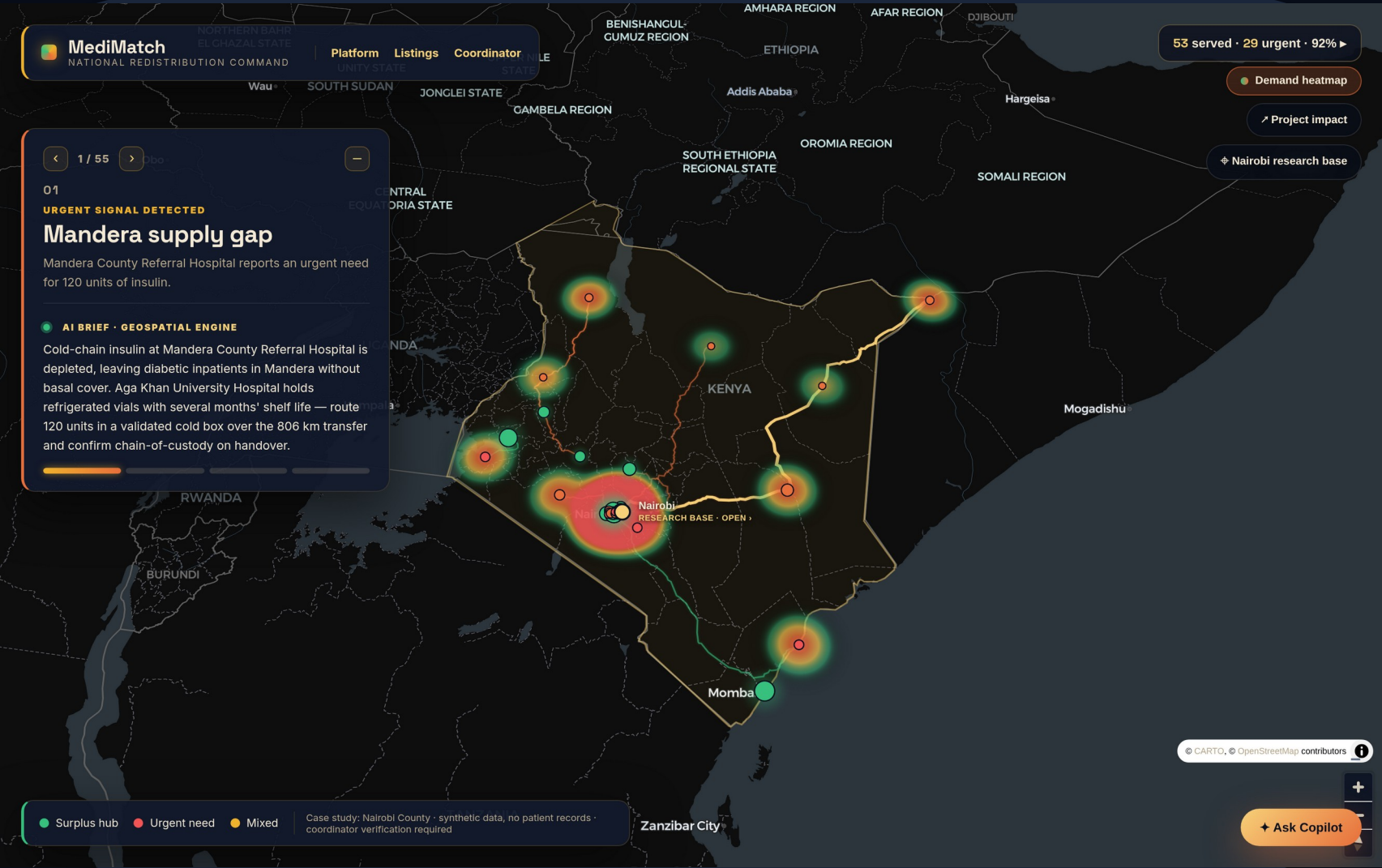


Q9 · 89.1% would pilot (Yes + Maybe)

From isolated reports to one national operating picture



See demand concentration before routing

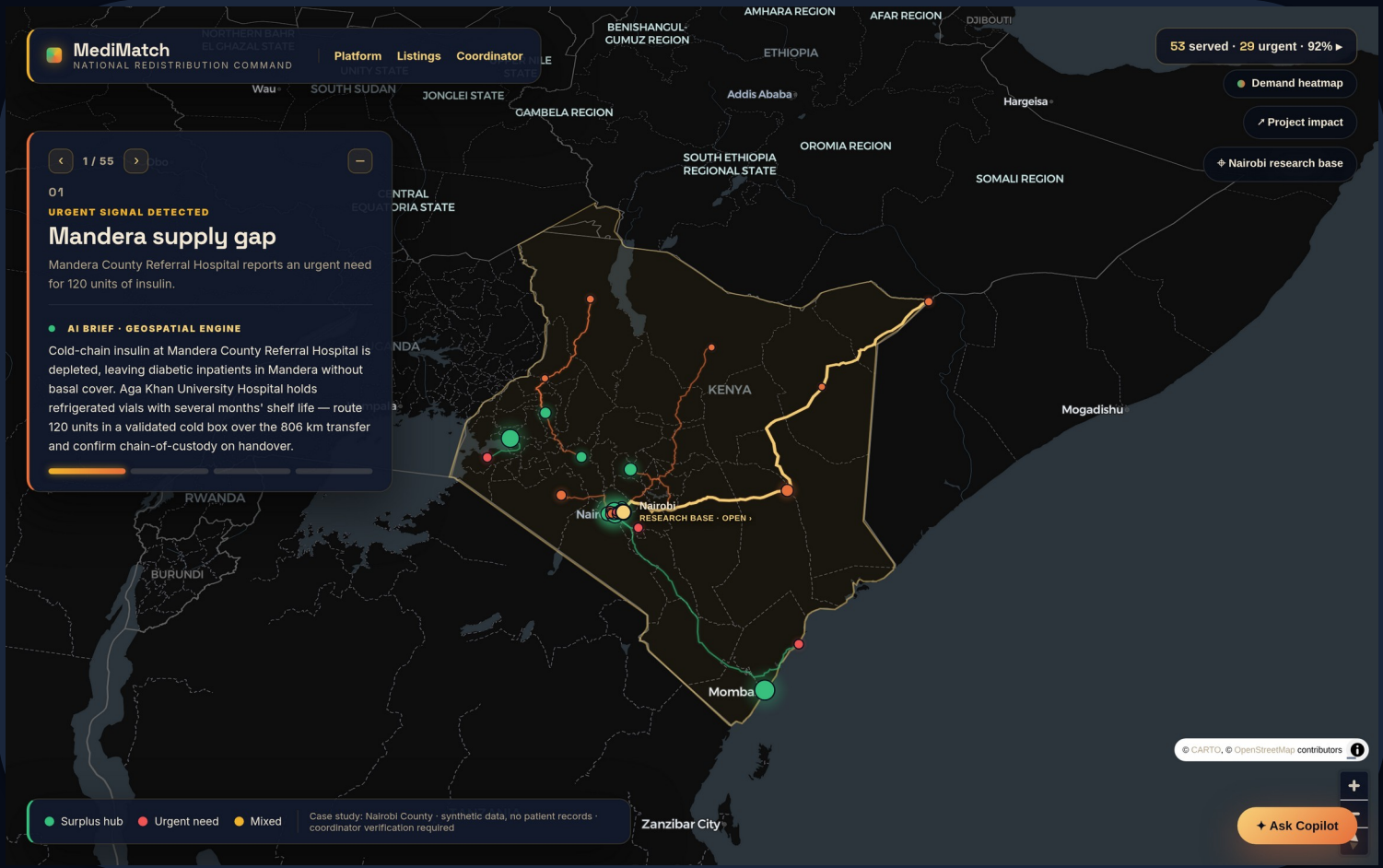


The heatmap reveals where shortfall clusters — read against hubs and road corridors. Distance is one signal among urgency, product fit and verification

Nairobi: the national model at street scale



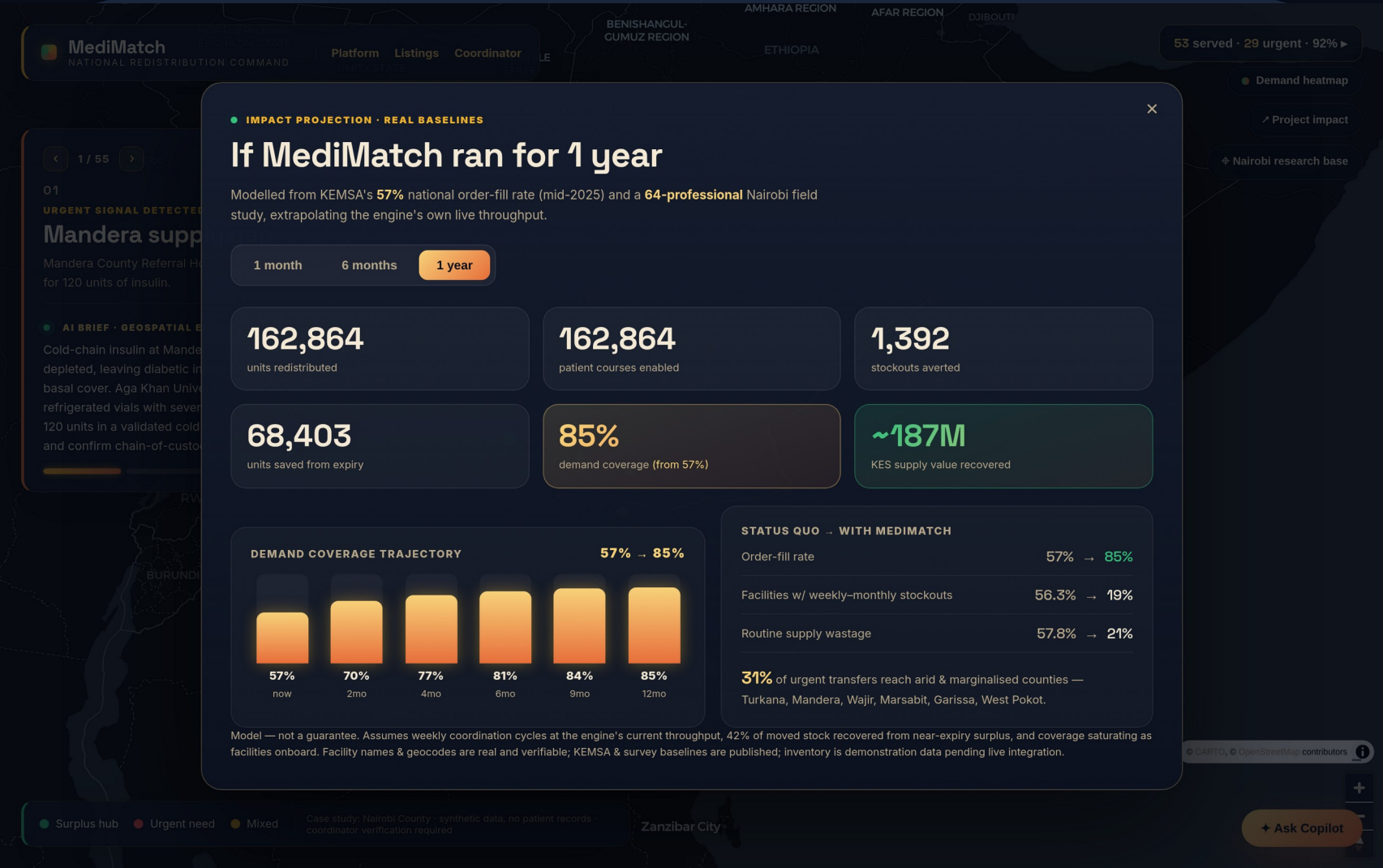
The system shows why a source was selected



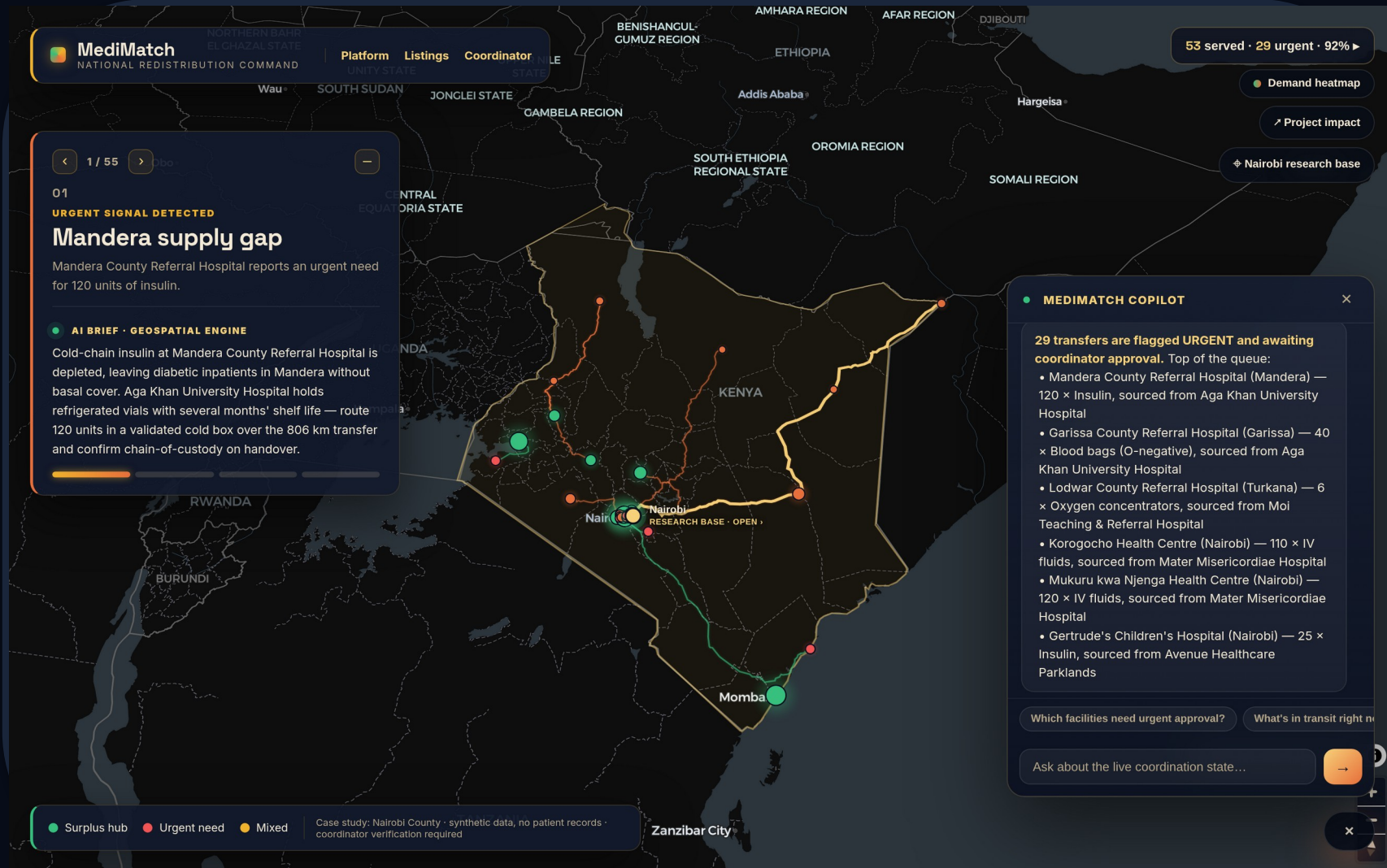
A plain-language situation brief accompanies every recommended route.

- Proximity
- Urgency
- Product fit
- Quantity
- Verification

What the current throughput could unlock over one year

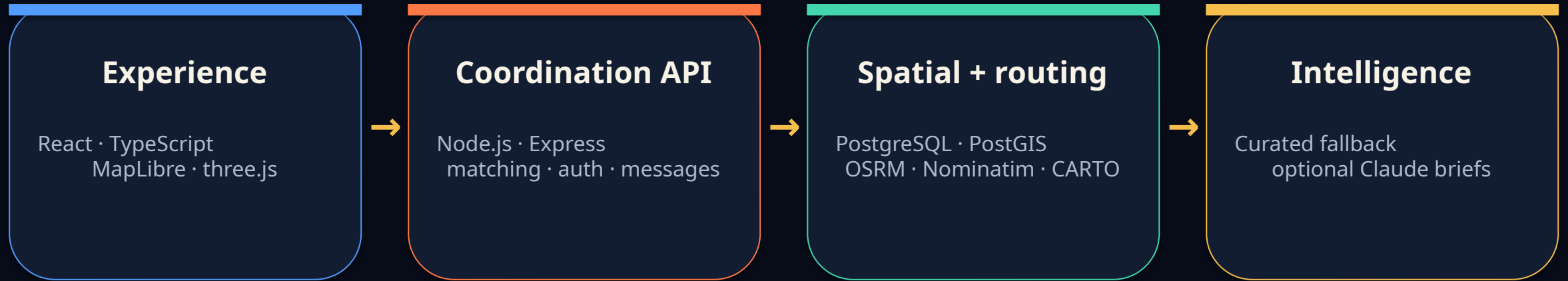


Coordination intelligence, grounded in live state



Answers drawn from the live coordination state — urgent approvals, in-transit transfers, county shortfalls, surplus hubs — not a general model

A production-minded spatial stack with a human decision boundary



Human verification is the final control

No diagnosis · no autonomous transfers · synthetic demo inventory · explicit assumptions · coordinator sign-off

Location is not just a field. It is a coordination advantage.

Evidence-based

64 healthcare professionals quantified the coordination gap.

Location-aware

Real-time, road-routed matching of surplus to urgent need.

Explainable

Transparent, multi-factor scoring — no black box.

Human-governed

Coordinator verification before every transfer.

MEDIMATCH

**See surplus. Detect need.
Coordinate impact.**

Thank you